

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Examination – 2023 Course: B. Tech. Branch: Mechanical & Allied Engg Semester: VI Subject Code & Name: BTMPE604D_Y22 Robotics Max Marks: 60 Date: 23-01-24 Duration: 3 Hr.			
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks	
Q. 1	Solve Any Two of the following.			12
A)	State and explain Cartesian and Cylindrical robot. Write their work envelope and applications.	CO1		6
B)	Define and explain any three Important specifications of an industrial robot.	CO1		6
C)	Write step by step solution of forward kinematics of a 3 DOF articulated robotic manipulator	CO2		6
Q.2	Solve Any Two of the following.			12
A)	What is robot Anatomy? Describe with neat sketch.	CO2		6
B)	Discuss about homogeneous transformation matrix and interprets the partitioning with examples.	CO2		6
C)	Discuss the types of drive systems used in robots	CO3		6
Q. 3	Solve Any Two of the following.			12
A)	Compare electrical, pneumatic & hydraulic drives for their characteristics.	CO3		6
B)	Explain steps for drive selection in robotic joints.	CO3		6
C)	Write any six sensors with their working principle and specific application in mobile robot for its navigation	CO4		6
Q.4	Solve Any Two of the following.			12
A)	Explain different types of proximity sensors with their applications in robotics.	CO4		6
B)	Explain different types of end effectors used in robotics.	CO4		6

C)	Explain the manual lead through and teach pendant programming in robot application	CO5	6
Q. 5	Solve Any Two of the following.		12
A)	Explain use of search techniques in robot problem solving by artificial intelligence.	CO5	6
B)	Explain use of robotics in assembly and inspection.	CO5	6
C)	Discuss in detail the selection and design considerations of grippers in robot.	CO4	6
	*** End ***		

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